

A thick dark blue vertical bar runs down the left side of the page. A medium blue arrow points to the right, overlapping the bar, with the date '1/25/2025' written inside it in white.

1/25/2025

Challenge Walkthrough

Internet Technology Extra Level

Rafael Tavares

554514

Several thin, curved lines in dark blue and light grey originate from the bottom left corner and sweep upwards and to the right, creating a dynamic, abstract design element.

Challenge Walkthrough

Level 1: Sending Bytes to the Server

Objective: I was instructed to send specific bytes to the server to continue the challenge.

What I Did:

- The server told me to send the bytes 0x50 0x39 0xa1.
- I simply sent them as raw bytes (not as characters or strings).
- After sending these bytes, I was able to move on to the next level.

Where I Struggled:

- There was no real struggle here. It was a simple task of sending the correct bytes.

Level 2: Sending the Student Number

Objective: I was asked to send my student number (as UTF-8 encoded bytes) to the server.

What I Did:

- The server asked for my student number, which was 554514.
- I sent the number in byte format, making sure it was encoded as UTF-8.

Where I Struggled:

- I initially thought there was some other format or encoding required, but it turned out UTF-8 was enough.

Level 3: Sending Octal Representation

Objective: The server sent some octal values and asked me to send them as bytes.

What I Did:

- The server gave me three octal values: 103, 247, and 231.
- I converted them to their byte equivalents (0x67, 0xA7, and 0xF1 respectively).
- I sent these as raw bytes and the server confirmed my input.

Where I Struggled:

- At first, I wasn't sure how to handle the octal numbers and thought I needed to do some complex conversion. It turns out, I just needed to treat them as octal byte values and send them directly.

Level 4: Base64 Decoding

Objective: I was given a Base64-encoded string and asked to decode it and send the raw output.

What I Did:

- The server provided a Base64-encoded string.
- I decoded it using `Base64.getDecoder().decode()`.
- After decoding, I sent the raw byte output to the server.

Where I Struggled:

- Initially, I missed that I had to send the raw bytes after decoding. I was trying to send the decoded string, not the raw byte data, but after a quick check, I fixed it and sent the right data.

Level 5: AES Decryption

Objective: The server gave me an AES-128 encrypted message and asked me to decrypt it.

What I Did:

- The server provided both the secret key and the ciphertext (both Base64-encoded).
- I decoded both using `Base64.getDecoder().decode()`.
- I used the AES decryption method with `Cipher.getInstance('AES/ECB/PKCS5Padding')` to decrypt the ciphertext.
- After decrypting the message, I sent it back to the server.

Where I Struggled:

- I struggled a bit with understanding the correct AES mode (ECB in this case) and ensuring I had the right padding scheme. But after testing a few different modes, I found that AES/ECB/PKCS5Padding was what worked.

Level 6: RSA Decryption

Objective: Decrypt an RSA-encrypted message using the provided private key and send the result back.

What I Did:

- Extracted the private key from the PEM format, decoded it, and converted it into a `PrivateKey` object.
- Cleaned and decoded the Base64 ciphertext into raw bytes.
- Decrypted the message using RSA/ECB/PKCS1Padding and sent the result to the server.

Where I Struggled:

- Figuring out how to handle the PEM-formatted private key.
- Fixing Base64 padding issues in the encrypted message.
- Identifying the correct RSA decryption mode and padding.

How I Overcame the Struggles:

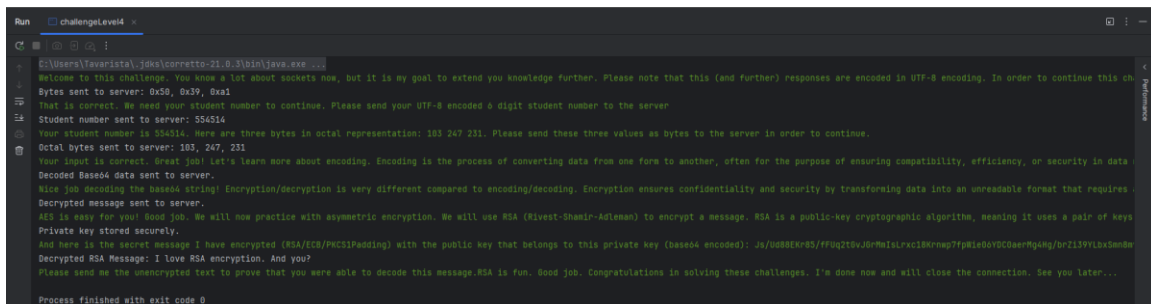
- Used Java's KeyFactory and PKCS8EncodedKeySpec to process the private key.
- Added logic to clean and pad the Base64 ciphertext.
- Experimented with decryption settings until RSA/ECB/PKCS1Padding worked.

Conclusion

Each level introduced a new concept in data encoding and encryption. While there were a few struggles along the way (especially with understanding some of the encodings and encryption methods), I was able to get through each challenge by carefully reading the server messages and adjusting my approach as I learned more.

Result

In here is the final result of the extra challenge.



```
Run challengeLevel4
C:\Users\Tavarista\jdk\corretto-21.0\bin>java.exe ...
Welcome to this challenge. You know a lot about sockets now, but it is my goal to extend you knowledge further. Please note that this (and further) responses are encoded in UTF-8 encoding. In order to continue this ch
Bytes sent to server: 0x50, 0x39, 0xa1
That is correct. We need your student number to continue. Please send your UTF-8 encoded 6 digit student number to the server.
Student number sent to server: 554514
Your student number is 554514. Here are three bytes in octal representation: 103 247 221. Please send these three values as bytes to the server in order to continue.
Octal bytes sent to server: 103, 247, 221
Your input is correct. Great job! Let's learn more about encoding. Encoding is the process of converting data from one form to another, often for the purpose of ensuring compatibility, efficiency, or security in data
Decoded Base64 data sent to server.
Nice job decoding the base64 string! Encryption/decryption is very different compared to encoding/decoding. Encryption ensures confidentiality and security by transforming data into an unreadable format that requires
Decrypted message sent to server.
AES is easy for you! Good job. We will now practice with asymmetric encryption. We will use RSA (Rivest-Shamir-Adleman) to encrypt a message. RSA is a public-key cryptographic algorithm, meaning it uses a pair of keys
Private key stored securely.
And here is the secret message I have encrypted (RSA/ECB/PKCS1Padding) with the public key that belongs to this private key (base64 encoded): J4U0BEXR05/FFQ2T0vDrM0IsIrc180rmq7FpW1a0Y0C0aerHyang/hr2139Ylbcxmdw
Decrypted RSA Message: I love RSA encryption. And you?
Please send me the unencrypted text to prove that you were able to decode this message. RSA is fun. Good job. Congratulations in solving these challenges. I'm done now and will close the connection. See you later...
Process finished with exit code 0
```